

C GOVERNING EQUATIONS IN MIXED-SPECTRAL FORMULATION

In this appendix, we give some more detail to the formulation of equations (2.15) and (2.16), as well as their matrix form implemented in PyFuga.

C.1 Fourier-transformed equations

We begin with the first-order momentum equation in its Cartesian form, eq. (2.9a),

$$U^0 \frac{\partial u_i^1}{\partial x} + \frac{\partial U^0}{\partial z} w^1 \delta_{i1} = \frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} - \frac{\partial p^1}{\partial x_i} + f_i. \quad (\text{C.1})$$

We can make use of the linearisation of the Navier-Stokes equations to simplify the handling of derivatives in x and y by applying a Fourier transform in the horizontal plane. Since the system of equations is linear, convolution in the wavenumber domain is avoided, and the spatial derivatives are replaced with products of the variables and their corresponding wavenumber. Let us analyse each term in the equation. The first term, the convective term, transforms as follows:

$$\mathcal{F}_{(x,y)} \left(U^0 \frac{\partial u_i^1}{\partial x} \right) = \mathcal{F}_{(x,y)} (U^0(z)) \cdot \mathcal{F}_{(x,y)} \left(\frac{\partial u_i^1}{\partial x} \right) = U^0 i k_1 \hat{u}_i^1. \quad (\text{C.2})$$

where $\hat{u}_i^1$ denotes the Fourier transform of u_i^1 , and k_1 is the wavenumber in the x -direction. Since U^0 is independent of x and y , it remains a function of z alone in the transformed equations. Here, $i k_1 \hat{u}_i^1$ arises from the Fourier transform of the derivative, which replaces $\frac{\partial}{\partial x}$ with $i k_1$. The pressure gradient term transforms differently depending on the direction i , such that

$$\mathcal{F}_{(x,y)} \left(-\frac{\partial p^1}{\partial x_i} \right) = \begin{cases} -i k_i \hat{p}^1 & \text{for } i = 1 \text{ or } 2, \\ -\frac{\partial p^1}{\partial z} & \text{for } i = 3. \end{cases} \quad (\text{C.3})$$

Note that, in the mixed-spectral formulation, it is often clearer to write out each equation explicitly rather than using Einstein notation, as $i = 1$ and $i = 2$ represent components in wavenumber space, while $i = 3$ remains in physical space, making it difficult to succinctly capture all dimensions in a single expression. The forcing term transforms straightforwardly to give

$$\mathcal{F}_{(x,y)} (f_i^1) = \hat{f}_i^1. \quad (\text{C.4})$$

Next, consider the second term on the left-hand side, the shear, or Coriolis, term, which transforms to give

$$\mathcal{F}_{(x,y)} \left(\frac{\partial U^0}{\partial z} w^1 \delta_{i1} \right) = \frac{\partial U^0}{\partial z} \hat{w}^1 \delta_{i1}. \quad (\text{C.5})$$

This term remains in the physical domain for z because $\partial U^0 / \partial z$ is independent of x and y . The Reynolds stress term is more complex and requires the use of the product rule. It can be written as

$$\begin{aligned} \mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) &= \mathcal{F}_{(x,y)} \left(\frac{\partial K}{\partial x_j} \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} + K \frac{\partial}{\partial x_j} \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) \\ &= \mathcal{F}_{(x,y)} \left(\frac{\partial K}{\partial x_j} \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} + K \left\{ \frac{\partial^2 u_i^1}{\partial x_j^2} + \frac{\partial^2 u_j^1}{\partial x_j \partial x_i} \right\} \right). \end{aligned} \quad (\text{C.6})$$

Note that for a simple eddy viscosity closure, such as that set out in section 2.1.2, $K = K(z)$ and thus cannot be taken outside of $\partial / \partial x_j$. With the equation in this form, we can apply the continuity equation for the first-order perturbation, i.e. divergence-free flow conditions $\partial u_j^1 / \partial x_j = 0$, to simplify it to

$$\begin{aligned} \mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) &= \mathcal{F}_{(x,y)} \left(\frac{\partial K}{\partial x_j} \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} + K \left\{ \frac{\partial^2 u_i^1}{\partial x_j^2} + \frac{\partial}{\partial x_i} \frac{\partial u_j^1}{\partial x_j} \right\} \right) \\ &= \mathcal{F}_{(x,y)} \left(\frac{\partial K}{\partial x_j} \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} + K \frac{\partial^2 u_i^1}{\partial x_j^2} \right). \end{aligned} \quad (\text{C.7})$$

Expanding the Einstein summation for the first term within the parentheses, and noting again that $K = K(z)$ is a function only of z , we can simplify eq. (C.7) to

$$\begin{aligned}
& \mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) \\
&= \mathcal{F}_{(x,y)} \left(\frac{\partial K}{\partial x} \left\{ \frac{\partial u_i^1}{\partial x} + \frac{\partial u^1}{\partial x_i} \right\} + \frac{\partial K}{\partial y} \left\{ \frac{\partial u_i^1}{\partial y} + \frac{\partial v^1}{\partial x_i} \right\} + \frac{\partial K}{\partial z} \left\{ \frac{\partial u_i^1}{\partial z} + \frac{\partial w^1}{\partial x_i} \right\} + K \frac{\partial^2 u_i^1}{\partial x_j^2} \right) \\
&= \mathcal{F}_{(x,y)} \left(\frac{dK}{dz} \left\{ \frac{\partial u_i^1}{\partial z} + \frac{\partial w^1}{\partial x_i} \right\} + K \frac{\partial^2 u_i^1}{\partial x_j^2} \right). \tag{C.8}
\end{aligned}$$

Lastly, we can expand the last term and apply the Fourier transform to arrive, for $i = 1$ or 2 , at

$$\begin{aligned}
& \mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial u_i^1}{\partial x_j} + \frac{\partial u_j^1}{\partial x_i} \right\} \right) \\
&= \mathcal{F}_{(x,y)} \left(\frac{\partial K}{\partial z} \left\{ \frac{\partial u_i^1}{\partial z} + \frac{\partial w^1}{\partial x_i} \right\} + K \left\{ \frac{\partial^2 u_i^1}{\partial x^2} + \frac{\partial^2 u_i^1}{\partial y^2} + \frac{\partial^2 u_i^1}{\partial z^2} \right\} \right) \\
&= \frac{\partial K}{\partial z} \left\{ \frac{\partial \hat{u}_i^1}{\partial z} + ik_i \hat{w}^1 \right\} + K \left\{ -(k_1^2 + k_2^2) \hat{u}_i^1 + \frac{\partial^2 \hat{u}_i^1}{\partial z^2} \right\} \\
&= -k^2 K \hat{u}_i^1 + \frac{\partial K}{\partial z} \frac{\partial \hat{u}_i^1}{\partial z} + K \frac{\partial^2 \hat{u}_i^1}{\partial z^2} + i \frac{\partial K}{\partial z} k_i \hat{w}^1 \\
&= -k^2 K \hat{u}_i^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{u}_i^1}{\partial z} \right) + i \frac{\partial K}{\partial z} k_i \hat{w}^1, \tag{C.9}
\end{aligned}$$

with k being the horizontal plane wave vector magnitude. For $i = 3$, we obtain

$$\mathcal{F}_{(x,y)} \left(\frac{\partial}{\partial x_j} K \left\{ \frac{\partial w^1}{\partial x_j} + \frac{\partial u_j^1}{\partial z} \right\} \right) = -k^2 K \hat{w}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{w}^1}{\partial z} \right) + \frac{\partial K}{\partial z} \frac{\partial \hat{w}^1}{\partial z}. \tag{C.10}$$

Applying the Fourier transform in x and y also to the first-order continuity equation (2.9b), we can now write the first-order system of equations in mixed-spectral form as

$$U^0 ik_1 \hat{u}^1 + \frac{\partial U^0}{\partial z} \hat{w}^1 = -k^2 K \hat{u}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{u}^1}{\partial z} \right) + \frac{dK}{dz} ik_1 \hat{w}^1 - ik_1 \hat{p}^1 + \hat{f}_1^1 \tag{C.11a}$$

$$U^0 ik_1 \hat{v}^1 = -k^2 K \hat{v}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{v}^1}{\partial z} \right) + \frac{dK}{dz} ik_2 \hat{w}^1 - ik_2 \hat{p}^1 \tag{C.11b}$$

$$U^0 ik_1 \hat{w}^1 = -k^2 K \hat{w}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{w}^1}{\partial z} \right) + \frac{dK}{dz} \frac{\partial \hat{w}^1}{\partial z} - \frac{\partial \hat{p}^1}{\partial z} \tag{C.11c}$$

$$ik_1 \hat{u}^1 + ik_2 \hat{v}^1 + \frac{\partial \hat{w}^1}{\partial z} = 0 \tag{C.11d}$$

with boundary conditions at the roughness and inversion height specified as

$$\begin{aligned}
\hat{u}_i^1(z_0) &= 0 \\
\hat{u}_i^1(z_i) &= 0. \tag{C.12}
\end{aligned}$$

Note that the boundary condition terms are unaffected by the transformation to the mixed-spectral domain. These boundary conditions are much more conveniently formulated in the spatial domain, which is one of the major reasons for keeping the Cartesian coordinate, z in the governing equations. Other reasons include the dependency of K on z for the simple closure scheme, which gives rise to terms that are non-linear in z , such as $K \frac{\partial \hat{w}^1}{\partial z}$, which would be less convenient to solve in the spectral domain.

C.2 Normalisation of the governing equations

In this section, we describe the steps to reformulate the governing equations as functions of non-dimensional variables. We will focus on the momentum and continuity equations from eqs. C.11. To simplify the representation of the horizontal wave vector, the first thing we do is to transform from a Cartesian coordinate system in (k_1, k_2) to the polar coordinates

$$k_1 = k \cos \beta, \quad k_2 = k \sin \beta, \quad (\text{C.13})$$

where k is the magnitude of the horizontal wavenumber, and β the phase angle. Substituting these definitions into eqs. C.11, we obtain:

$$\begin{aligned} U^0 ik \cos \beta \hat{u}^1 + \frac{\partial U^0}{\partial z} \hat{w}^1 &= -k^2 K \hat{u}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{u}^1}{\partial z} \right) + \frac{\partial K}{\partial z} ik \cos \beta \hat{w}^1 - ik \cos \beta \hat{p}^1 + \hat{f}_1^1 \\ U^0 ik \cos \beta \hat{v}^1 &= -k^2 K \hat{v}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{v}^1}{\partial z} \right) + \frac{\partial K}{\partial z} ik \sin \beta \hat{w}^1 - ik \sin \beta \hat{p}^1 \\ U^0 ik \cos \beta \hat{w}^1 &= -k^2 K \hat{w}^1 + \frac{\partial}{\partial z} (K (-ik \cos \beta \hat{u}^1 - ik \sin \beta \hat{v}^1)) \\ &\quad + \frac{\partial K}{\partial z} (-ik \cos \beta \hat{u}^1 - ik \sin \beta \hat{v}^1) - \frac{\partial \hat{p}^1}{\partial z}, \end{aligned} \quad (\text{C.14})$$

where $\frac{\partial \hat{w}^1}{\partial z}$ has been eliminated from eq. (C.11c) using the continuity equation,

$$ik \cos \beta \hat{u}^1 + ik \sin \beta \hat{v}^1 + \frac{\partial \hat{w}^1}{\partial z} = 0. \quad (\text{C.15})$$

The second item to address is the forcing term. The forcing terms in the system of equations, $\hat{f}_i^1(\mathbf{k}, z)$, cannot generally be separated into independent functions of $\mathbf{k} = (k_x, k_y)$ and z . Consider the Fourier transform of the infinitely thin actuator disc defined in eq. (2.6), $f_i(x, y, z)$, we have

$$\begin{aligned} \hat{f}_i(k_x, k_y, z) &= \frac{1}{(2\pi)^2} e^{-ix_h} \int_{-\infty}^{\infty} dy f(x_h, y, z) e^{-iyk_y} \\ &= \frac{1}{(2\pi)^2} e^{-ix_h} \int_{-\infty}^{\infty} dy \left(-\frac{1}{2} C_t U_{\text{free}}^2 \Theta(R^2 - (y - y_h)^2 - (z - z_h)^2) \right) e^{-iyk_y}. \end{aligned} \quad (\text{C.16})$$

where $\Theta(\cdot)$ is the Heaviside step function. In this integral, the complexity arises because the freestream velocity, U_{free} , is a function of z , which is then multiplied by the Heaviside function, which itself is a function of both y and z . As a result, when we take the Fourier transform in the z -direction, this mixed dependence prevents the forcing term from being expressed as a product of independent functions. As shown in section 2.2.1, the forcing term $\hat{f}_i^1(\mathbf{k}, z)$ is, therefore, approximated as a sum of products:

$$\hat{f}_i^1(\mathbf{k}, z) = \sum_n f_{n,\mathbf{k}}^i \phi_n(z), \quad (\text{C.17})$$

where $f_{n,\mathbf{k}}^i$ captures the dependence on $\mathbf{k}$, including details of the thrust coefficient and wind turbine geometry, while $\phi_n(z)$ is a set of triangular basis functions defined in eq. (2.14). Moreover, in eq. (C.14) we only show $\hat{f}_1^1(\mathbf{k}, z)$, the longitudinal forcing term, because any lateral and vertical components (resulting from yaw or tilt) can be solved independently and then combined to yield the complete solution. This approach is possible due to the linearity of Fuga.

Returning to the first-order system of equations in eq. (C.14), we can now write that for a forcing applied at a vertical level n ,

$$\begin{aligned} U^0 ik \cos \beta \hat{u}^1 + \frac{\partial U^0}{\partial z} \hat{w}^1 &= -k^2 K \hat{u}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{u}^1}{\partial z} \right) + \frac{\partial K}{\partial z} ik \cos \beta \hat{w}^1 - ik \cos \beta \hat{p}^1 + f_{n,\mathbf{k}}^1 \phi_n \\ U^0 ik \cos \beta \hat{v}^1 &= -k^2 K \hat{v}^1 + \frac{\partial}{\partial z} \left(K \frac{\partial \hat{v}^1}{\partial z} \right) + \frac{\partial K}{\partial z} ik \sin \beta \hat{w}^1 - ik \sin \beta \hat{p}^1 \\ U^0 ik \cos \beta \hat{w}^1 &= -k^2 K \hat{w}^1 + \frac{\partial}{\partial z} (K (-ik \cos \beta \hat{u}^1 - ik \sin \beta \hat{v}^1)) \\ &\quad + \frac{\partial K}{\partial z} (-ik \cos \beta \hat{u}^1 - ik \sin \beta \hat{v}^1) - \frac{\partial \hat{p}^1}{\partial z}. \end{aligned} \quad (\text{C.18})$$

To express this system of equations in a non-dimensional form, we start by introducing the following new dependent variables:

$$\tilde{u} = \frac{\hat{u}^1 u_* k}{f_{n,k}^1}, \quad \tilde{v} = \frac{\hat{v}^1 u_* k}{f_{n,k}^1}, \quad \tilde{w} = \frac{\hat{w}^1 u_* k}{f_{n,k}^1}, \quad \text{and} \quad \tilde{p} = \frac{\hat{p}^1 k}{f_{n,k}^1}. \quad (\text{C.19})$$

Substituting these into eqs. (C.15) and (C.18), we obtain

$$\begin{aligned} U^0 i k \cos \beta \tilde{u} \frac{f_{n,k}^1}{u_* k} + \frac{\partial U^0}{\partial z} \tilde{w} \frac{f_{n,k}^1}{u_* k} &= -k^2 K \tilde{u} \frac{f_{n,k}^1}{u_* k} + \frac{\partial}{\partial z} \left(K \frac{\partial \tilde{u}}{\partial z} \frac{f_{n,k}^1}{u_* k} \right) + \frac{\partial K}{\partial z} i k \cos \beta \tilde{w} \frac{f_{n,k}^1}{u_* k} \\ &\quad - i k \cos \beta \tilde{p} \frac{f_{n,k}^1}{k} + f_{n,k}^1 \phi_n \\ U^0 i k \cos \beta \tilde{v} \frac{f_{n,k}^1}{u_* k} &= -k^2 K \tilde{v} \frac{f_{n,k}^1}{u_* k} + \frac{\partial}{\partial z} \left(K \frac{\partial \tilde{v}}{\partial z} \frac{f_{n,k}^1}{u_* k} \right) + \frac{\partial K}{\partial z} i k \sin \beta \tilde{w} \frac{f_{n,k}^1}{u_* k} \\ &\quad - i k \sin \beta \tilde{p} \frac{f_{n,k}^1}{k} \\ U^0 i k \cos \beta \tilde{w} \frac{f_{n,k}^1}{u_* k} &= -k^2 K \tilde{w} \frac{f_{n,k}^1}{u_* k} + \frac{\partial}{\partial z} \left(K \left(-i k \cos \beta \tilde{u} \frac{f_{n,k}^1}{u_* k} - i k \sin \beta \tilde{v} \frac{f_{n,k}^1}{u_* k} \right) \right) \\ &\quad + \frac{\partial K}{\partial z} \left(-i k \cos \beta \tilde{u} \frac{f_{n,k}^1}{u_* k} - i k \sin \beta \tilde{v} \frac{f_{n,k}^1}{u_* k} \right) - \frac{\partial \tilde{p}}{\partial z} \frac{f_{n,k}^1}{k} \\ i k \cos \beta \tilde{u} \frac{f_{n,k}^1}{u_* k} + i k \sin \beta \tilde{v} \frac{f_{n,k}^1}{u_* k} + \frac{\partial \tilde{w}}{\partial z} \frac{f_{n,k}^1}{u_* k} &= 0, \end{aligned} \quad (\text{C.20})$$

which can be simplified and written as

$$\begin{aligned} \frac{U^0}{u_*} i k \cos \beta \tilde{u} + \frac{1}{k} \frac{\partial}{\partial z} \left(\frac{U_0}{u_*} \right) \tilde{w} &= -\frac{kK}{u_*} \tilde{u} + \frac{1}{k^2} \frac{\partial}{\partial z} \left(\frac{kK}{u_*} \frac{\partial \tilde{u}}{\partial z} \right) + \frac{1}{k} \frac{\partial}{\partial z} \left(\frac{kK}{u_*} \right) i k \cos \beta \tilde{w} - i k \cos \beta \tilde{p} + \phi_n \\ \frac{U^0}{u_*} i k \cos \beta \tilde{v} &= -\frac{kK}{u_*} \tilde{v} + \frac{1}{k^2} \frac{\partial}{\partial z} \left(\frac{kK}{u_*} \frac{\partial \tilde{v}}{\partial z} \right) + \frac{1}{k} \frac{\partial}{\partial z} \left(\frac{kK}{u_*} \right) i k \sin \beta \tilde{w} - i k \cos \beta \tilde{p} \\ \frac{U^0}{u_*} i k \cos \beta \tilde{w} &= -\frac{kK}{u_*} \tilde{w} + \frac{1}{k} \frac{\partial}{\partial z} \left(\frac{kK}{u_*} (-i k \cos \beta \tilde{u} - i k \sin \beta \tilde{v}) \right) \\ &\quad + \frac{1}{k} \frac{\partial}{\partial z} \left(\frac{kK}{u_*} \right) (-i k \cos \beta \tilde{u} - i k \sin \beta \tilde{v}) - \frac{1}{k} \frac{\partial \tilde{p}}{\partial z} \\ i k \cos \beta \tilde{u} + i k \sin \beta \tilde{v} + \frac{1}{k} \frac{\partial \tilde{w}}{\partial z} &= 0. \end{aligned} \quad (\text{C.21})$$

Writing the system of equations this way allows us to substitute two new non-dimensional variables,

$$U^{0*} \equiv \frac{U^0}{u_*} \quad \text{and} \quad K^* \equiv \frac{kK}{u_*}, \quad (\text{C.22})$$

to obtain

$$\begin{aligned} U^{0*} i k \cos \beta \tilde{u} + \frac{1}{k} \frac{\partial U^{0*}}{\partial z} \tilde{w} &= -K^* \tilde{u} + \frac{1}{k^2} \frac{\partial}{\partial z} \left(K^* \frac{\partial \tilde{u}}{\partial z} \right) + \frac{1}{k} \frac{\partial K^*}{\partial z} i k \cos \beta \tilde{w} - i k \cos \beta \tilde{p} + \phi_n \\ U^{0*} i k \cos \beta \tilde{v} &= -K^* \tilde{v} + \frac{1}{k^2} \frac{\partial}{\partial z} \left(K^* \frac{\partial \tilde{v}}{\partial z} \right) + \frac{1}{k} \frac{\partial K^*}{\partial z} i k \sin \beta \tilde{w} - i k \cos \beta \tilde{p} \\ U^{0*} i k \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{1}{k} \frac{\partial}{\partial z} \left(K^* (-i k \cos \beta \tilde{u} - i k \sin \beta \tilde{v}) \right) \\ &\quad + \frac{1}{k} \frac{\partial K^*}{\partial z} (-i k \cos \beta \tilde{u} - i k \sin \beta \tilde{v}) - \frac{1}{k} \frac{\partial \tilde{p}}{\partial z} \\ i k \cos \beta \tilde{u} + i k \sin \beta \tilde{v} + \frac{1}{k} \frac{\partial \tilde{w}}{\partial z} &= 0. \end{aligned} \quad (\text{C.23})$$

To complete the normalise this system of equations, we also need to non-dimensionalise the vertical coordinate, z . We can do this by introducing another variable $s = kz$, with $ds = kdz$. Substituting this and simplifying leads to:

$$\begin{aligned}
U^{0*} i \cos \beta \tilde{u} + \frac{\partial U^{0*}}{\partial s} \tilde{w} &= -K^* \tilde{u} + \frac{\partial}{\partial s} \left(K^* \frac{\partial \tilde{u}}{\partial s} \right) + \frac{\partial K^*}{\partial s} i \cos \beta \tilde{w} - i \cos \beta \tilde{p} + \phi_n \\
U^{0*} i \cos \beta \tilde{v} &= -K^* \tilde{v} + \frac{\partial}{\partial s} \left(K^* \frac{\partial \tilde{v}}{\partial s} \right) + \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} - i \sin \beta \tilde{p} \\
U^{0*} i \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{\partial}{\partial s} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) \\
&\quad + \frac{\partial K^*}{\partial s} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) - \frac{\partial \tilde{p}}{\partial s} \\
i \cos \beta \tilde{u} + i \sin \beta \tilde{v} + \frac{\partial \tilde{w}}{\partial s} &= 0.
\end{aligned} \tag{C.24}$$

The last item to address is the simplification of the term $\frac{\partial U^{0*}}{\partial s} \tilde{w}$ in the x -momentum equation. Starting with the definition of the Monin-Obukhov velocity profile used in equation (2.4)

$$\frac{\kappa z}{u_*} \frac{\partial U^0}{\partial z} = \phi_m, \tag{C.25}$$

we can substitute the definitions $U^0 = u_* U^{0*}$ and $\frac{\partial}{\partial z} = k \frac{\partial}{\partial s}$ to give

$$\frac{k \kappa z}{u_*} \frac{\partial U^{0*}}{\partial s} = \phi_m \quad \Rightarrow \quad k \kappa z \frac{\partial U^{0*}}{\partial s} = \phi_m. \tag{C.26}$$

Rearranging this and introducing u_* as

$$\frac{\partial U^{0*}}{\partial s} = \frac{1}{k} \frac{\phi_m}{\kappa z} \quad \Rightarrow \quad \frac{\partial U^{0*}}{\partial s} = \frac{1}{k} \frac{u_*}{\left(\frac{\kappa u_*^* z}{\phi_m} \right)}, \tag{C.27}$$

allows us to substitute the eddy viscosity from equation (2.4) and apply the definition $K \equiv \frac{u_* K^*}{k}$ such that

$$\frac{\partial U^{0*}}{\partial s} = \frac{u_*}{kK} \quad \Rightarrow \quad \frac{\partial U^{0*}}{\partial s} = \frac{1}{K^*}. \tag{C.28}$$

Applying this simplification gives us the complete non-dimensionalised system of equations:

$$\begin{aligned}
U^{0*} i \cos \beta \tilde{u} + \frac{1}{K^*} \tilde{w} &= -K^* \tilde{u} + \frac{\partial}{\partial s} \left(K^* \frac{\partial \tilde{u}}{\partial s} \right) + \frac{\partial K^*}{\partial s} i \cos \beta \tilde{w} - i \cos \beta \tilde{p} + \phi_n \\
U^{0*} i \cos \beta \tilde{v} &= -K^* \tilde{v} + \frac{\partial}{\partial s} \left(K^* \frac{\partial \tilde{v}}{\partial s} \right) + \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} - i \sin \beta \tilde{p} \\
U^{0*} i \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{\partial}{\partial s} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) \\
&\quad + \frac{\partial K^*}{\partial s} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) - \frac{\partial \tilde{p}}{\partial s} \\
i \cos \beta \tilde{u} + i \sin \beta \tilde{v} + \frac{\partial \tilde{w}}{\partial s} &= 0.
\end{aligned} \tag{C.29}$$

C.3 Governing equations as a system of ODEs

We now want to rewrite these as a system of ordinary differential equations in six unknowns, $\{\tilde{u}, \tilde{u}', \tilde{v}, \tilde{v}', \tilde{w}, \tilde{p}\}$, where we have defined two new variables,

$$\tilde{u}' = \frac{\partial \tilde{u}}{\partial s} \quad \text{and} \quad \tilde{v}' = \frac{\partial \tilde{v}}{\partial s}, \tag{C.30}$$

to simplify the structure of the equations and reduce the number of higher-order derivatives. We can immediately extract an equation for $\frac{\partial \tilde{w}}{\partial s}$ by rearranging the continuity equation from equation (C.29) as

$$\frac{\partial \tilde{w}}{\partial s} = -i \cos \beta \tilde{u} - i \sin \beta \tilde{v}. \quad (\text{C.31})$$

Substituting the new variables $\tilde{u}'$ and $\tilde{v}'$ into equation (C.29) and dividing both sides of the u - and v -momentum equations by K^* we get

$$\begin{aligned} \frac{U^{0*} i \cos \beta \tilde{u}}{K^*} + \frac{1}{K^{*2}} \tilde{w} &= -\tilde{u} + \frac{1}{K^*} \frac{\partial}{\partial s} (K^* \tilde{u}') + \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \cos \beta \tilde{w} - \frac{i \cos \beta \tilde{p}}{K^*} + \frac{\phi_n}{K^*} \\ \frac{U^{0*} i \cos \beta \tilde{v}}{K^*} &= -\tilde{v} + \frac{1}{K^*} \frac{\partial}{\partial s} (K^* \tilde{v}') + \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} - \frac{i \sin \beta \tilde{p}}{K^*} \\ U^{0*} i \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{\partial}{\partial s} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) \\ &\quad + \frac{\partial K^*}{\partial s} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) - \frac{\partial \tilde{p}}{\partial s}. \end{aligned} \quad (\text{C.32})$$

We can now rewrite the diffusion terms, $\frac{\partial}{\partial s} (K^* (\cdot))$, using the product rule. For the x - and y -momentum equations, this yields

$$\begin{aligned} \frac{U^{0*} i \cos \beta \tilde{u}}{K^*} + \frac{1}{K^{*2}} \tilde{w} &= -\tilde{u} + \frac{\partial \tilde{u}'}{\partial s} + \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{u}' + \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \cos \beta \tilde{w} - \frac{i \cos \beta \tilde{p}}{K^*} + \frac{\phi_n}{K^*} \\ \frac{U^{0*} i \cos \beta \tilde{v}}{K^*} &= -\tilde{v} + \frac{\partial \tilde{v}'}{\partial s} + \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{v}' + \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} - \frac{i \sin \beta \tilde{p}}{K^*}. \end{aligned} \quad (\text{C.33})$$

The z -momentum equation is, however, slightly more complex. Focusing only on the diffusion terms, we can show that

$$\begin{aligned} &\frac{\partial}{\partial s} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) + \frac{\partial K^*}{\partial s} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) \\ &= -i \cos \beta \frac{\partial}{\partial s} (K^* \tilde{u}) - i \sin \beta \frac{\partial}{\partial s} (K^* \tilde{v}) - i \cos \beta \frac{\partial K^*}{\partial s} \tilde{u} - i \sin \beta \frac{\partial K^*}{\partial s} \tilde{v} \\ &= -i \cos \beta \left(K^* \tilde{u}' + \frac{\partial K^*}{\partial s} \tilde{u} \right) - i \sin \beta \left(K^* \tilde{v}' + \frac{\partial K^*}{\partial s} \tilde{v} \right) - i \cos \beta \frac{\partial K^*}{\partial s} \tilde{u} - i \sin \beta \frac{\partial K^*}{\partial s} \tilde{v} \\ &= -2i \cos \beta \frac{\partial K^*}{\partial s} \tilde{u} - i K^* \cos \beta \tilde{u}' - 2i \sin \beta \frac{\partial K^*}{\partial s} \tilde{v} - i K^* \sin \beta \tilde{v}'. \end{aligned} \quad (\text{C.34})$$

The complete z -momentum equation is then

$$U^{0*} i \cos \beta \tilde{w} = -K^* \tilde{w} - 2i \cos \beta \frac{\partial K^*}{\partial s} \tilde{u} - i K^* \cos \beta \tilde{u}' - 2i \sin \beta \frac{\partial K^*}{\partial s} \tilde{v} - i K^* \sin \beta \tilde{v}' - \frac{\partial \tilde{p}}{\partial s}. \quad (\text{C.35})$$

These three momentum equations can be rearranged for $\frac{\partial \tilde{u}'}{\partial s}$, $\frac{\partial \tilde{v}'}{\partial s}$, and $\frac{\partial \tilde{p}}{\partial s}$, respectively,

$$\begin{aligned} \frac{\partial \tilde{u}'}{\partial s} &= \tilde{u} + \frac{U^{0*} i \cos \beta \tilde{u}}{K^*} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{u}' + \frac{1}{K^{*2}} \tilde{w} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \cos \beta \tilde{w} + \frac{i \cos \beta \tilde{p}}{K^*} - \frac{\phi_n}{K^*} \\ \frac{\partial \tilde{v}'}{\partial s} &= \tilde{v} + \frac{U^{0*} i \cos \beta \tilde{v}}{K^*} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{v}' - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} + \frac{i \sin \beta \tilde{p}}{K^*} \\ \frac{\partial \tilde{p}}{\partial s} &= -2i \cos \beta \frac{\partial K^*}{\partial s} \tilde{u} - i K^* \cos \beta \tilde{u}' - 2i \sin \beta \frac{\partial K^*}{\partial s} \tilde{v} - i K^* \sin \beta \tilde{v}' - K^* \tilde{w} - U^{0*} i \cos \beta \tilde{w}. \end{aligned} \quad (\text{C.36})$$

Applying this simplification and some refactoring, we can now write the full system of equations in terms of the six unknowns, $\{\tilde{u}, \tilde{u}', \tilde{v}, \tilde{v}', \tilde{w}, \tilde{p}\}$, as

$$\begin{aligned}
\frac{\partial \tilde{u}}{\partial s} &= \tilde{u}' \\
\frac{\partial \tilde{u}'}{\partial s} &= \left(1 + \frac{i \cos \beta U^{0*}}{K^*}\right) \tilde{u} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{u}' + \left(\frac{1}{K^{*2}} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \cos \beta\right) \tilde{w} + \frac{i \cos \beta}{K^*} \tilde{p} - \frac{\phi_n}{K^*} \\
\frac{\partial \tilde{v}}{\partial s} &= \tilde{v}' \\
\frac{\partial \tilde{v}'}{\partial s} &= \left(1 + \frac{i \cos \beta U^{0*}}{K^*}\right) \tilde{v} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} \tilde{v}' - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \sin \beta \tilde{w} + \frac{i \sin \beta}{K^*} \tilde{p} \\
\frac{\partial \tilde{w}}{\partial s} &= -i \cos \beta \tilde{u} - i \sin \beta \tilde{v} \\
\frac{\partial \tilde{p}}{\partial s} &= -2i \cos \beta \frac{\partial K^*}{\partial s} \tilde{u} - i K^* \cos \beta \tilde{u}' - 2i \sin \beta \frac{\partial K^*}{\partial s} \tilde{v} - i K^* \sin \beta \tilde{v}' - (K^* + U^{0*} i \cos \beta) \tilde{w}.
\end{aligned} \tag{C.37}$$

As mentioned in section 2.2.1, the presence of $K^*(s)$ in the denominator of many of the terms in eq. (C.37) means that these equations become numerically unstable for small values of s , i.e. close to the ground. In this case, instead of s we use the independent variable

$$t = U^{0*} \kappa, \quad \text{with} \quad \frac{\partial t}{\partial z} = \frac{\partial U^{0*}}{\partial z} \kappa = \frac{\kappa k}{K^*}. \tag{C.38}$$

The transition point between s and t is defined unambiguously as the point satisfying $\frac{\partial}{\partial s} = \frac{\partial}{\partial t}$ (full details are given in appendix B). The system of equations can be derived directly from eq. (C.37) by providing a transformation between s and t using the definition of t from eq. (C.38) and the simplification of $\frac{\partial U^{0*}}{\partial s}$ from eq. (C.28) as

$$\frac{\partial}{\partial s} = \frac{\partial t}{\partial s} \frac{\partial}{\partial t} = \kappa \frac{\partial U^{0*}}{\partial s} \frac{\partial}{\partial t} = \frac{\kappa}{K^*} \frac{\partial}{\partial t}. \tag{C.39}$$

Substituting this directly into (C.37) leads to a rather messy formulation. It is clearer to instead apply it to (C.29), which gives

$$\begin{aligned}
U^{0*} i \cos \beta \tilde{u} + \frac{1}{K^*} \tilde{w} &= -K^* \tilde{u} + \frac{\kappa}{K^*} \frac{\partial}{\partial t} \left(K^* \frac{\kappa}{K^*} \frac{\partial \tilde{u}}{\partial t} \right) + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} i \cos \beta \tilde{w} - i \cos \beta \tilde{p} + \phi_n \\
U^{0*} i \cos \beta \tilde{v} &= -K^* \tilde{v} + \frac{\kappa}{K^*} \frac{\partial}{\partial t} \left(K^* \frac{\kappa}{K^*} \frac{\partial \tilde{v}}{\partial t} \right) + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} i \sin \beta \tilde{w} - i \sin \beta \tilde{p} \\
U^{0*} i \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{\kappa}{K^*} \frac{\partial}{\partial t} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) \\
&\quad + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) - \frac{\kappa}{K^*} \frac{\partial \tilde{p}}{\partial t} \\
i \cos \beta \tilde{u} + i \sin \beta \tilde{v} + \frac{\kappa}{K^*} \frac{\partial \tilde{w}}{\partial t} &= 0.
\end{aligned} \tag{C.40}$$

We can focus on the momentum equations by rearranging the continuity equation to arrive immediately at

$$\frac{\partial \tilde{w}}{\partial t} = -i \frac{K^*}{\kappa} \cos \beta \tilde{u} - i \frac{K^*}{\kappa} \sin \beta \tilde{v}. \tag{C.41}$$

Since κ is a constant, the momentum equations can be simplified to yield

$$\begin{aligned}
U^{0*} i \cos \beta \tilde{u} + \frac{1}{K^*} \tilde{w} &= -K^* \tilde{u} + \frac{\kappa^2}{K^*} \frac{\partial}{\partial t} \left(\frac{\partial \tilde{u}}{\partial t} \right) + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} i \cos \beta \tilde{w} - i \cos \beta \tilde{p} + \phi_n \\
U^{0*} i \cos \beta \tilde{v} &= -K^* \tilde{v} + \frac{\kappa^2}{K^*} \frac{\partial}{\partial t} \left(\frac{\partial \tilde{v}}{\partial t} \right) + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} i \sin \beta \tilde{w} - i \sin \beta \tilde{p} \\
U^{0*} i \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{\kappa}{K^*} \frac{\partial}{\partial t} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) \\
&\quad + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) - \frac{\kappa}{K^*} \frac{\partial \tilde{p}}{\partial t}.
\end{aligned} \tag{C.42}$$

This makes evident one of the advantages of swapping s for t : the convection term no longer requires the application of the product rule, and thus we do not get a term in $\tilde{u}'$ on the right-hand side of the x -momentum equation, nor in $\tilde{v}'$ in the y -momentum equation. If we now define the independent variables

$$\tilde{u}' = \frac{\partial \tilde{u}}{\partial t} \quad \text{and} \quad \tilde{v}' = \frac{\partial \tilde{v}}{\partial t}, \quad (\text{C.43})$$

and substitute them into the momentum equations, the following governing equations result:

$$\begin{aligned} U^{0*} i \cos \beta \tilde{u} + \frac{1}{K^*} \tilde{w} &= -K^* \tilde{u} + \frac{\kappa^2}{K^*} \frac{\partial \tilde{u}'}{\partial t} + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} i \cos \beta \tilde{w} - i \cos \beta \tilde{p} + \phi_n \\ U^{0*} i \cos \beta \tilde{v} &= -K^* \tilde{v} + \frac{\kappa^2}{K^*} \frac{\partial \tilde{v}'}{\partial t} + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} i \sin \beta \tilde{w} - i \sin \beta \tilde{p} \\ U^{0*} i \cos \beta \tilde{w} &= -K^* \tilde{w} + \frac{\kappa}{K^*} \frac{\partial}{\partial t} (K^* (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v})) \\ &\quad + \frac{\kappa}{K^*} \frac{\partial K^*}{\partial t} (-i \cos \beta \tilde{u} - i \sin \beta \tilde{v}) - \frac{\kappa}{K^*} \frac{\partial \tilde{p}}{\partial t}. \end{aligned} \quad (\text{C.44})$$

We can now rearrange eq. (C.44) as ordinary differential equations analogous to eq. (C.37), but formulated in the t vertical coordinate. The whole system of equations is then written as

$$\begin{aligned} \frac{\partial \tilde{u}}{\partial t} &= \tilde{u}' \\ \frac{\partial \tilde{u}'}{\partial t} &= \left(\frac{K^{*2}}{\kappa^2} + \frac{i K^* \cos \beta U^{0*}}{\kappa^2} \right) \tilde{u} + \left(\frac{1}{\kappa^2} - \frac{1}{\kappa} \frac{\partial K^*}{\partial t} i \cos \beta \right) \tilde{w} + \frac{i K^* \cos \beta}{\kappa^2} \tilde{p} - \frac{K^* \phi_n}{\kappa^2} \\ \frac{\partial \tilde{v}}{\partial t} &= \tilde{v}' \\ \frac{\partial \tilde{v}'}{\partial t} &= \left(\frac{K^{*2}}{\kappa^2} + \frac{i K^* \cos \beta U^{0*}}{\kappa^2} \right) \tilde{v} - \frac{1}{\kappa} \frac{\partial K^*}{\partial t} i \sin \beta \tilde{w} + \frac{i K^* \sin \beta}{\kappa^2} \tilde{p} \\ \frac{\partial \tilde{w}}{\partial t} &= -i \frac{K^*}{\kappa} \cos \beta \tilde{u} - i \frac{K^*}{\kappa} \sin \beta \tilde{v} \\ \frac{\partial \tilde{p}}{\partial t} &= -2i \cos \beta \frac{\partial K^*}{\partial t} \tilde{u} - i K^* \cos \beta \tilde{u}' - 2i \sin \beta \frac{\partial K^*}{\partial t} \tilde{v} - i K^* \sin \beta \tilde{v}' - \left(\frac{K^{*2}}{\kappa} + \frac{K^*}{\kappa} U^{0*} i \cos \beta \right) \tilde{w}. \end{aligned} \quad (\text{C.45})$$

C.4 Matrix formulation of the governing equations

Equations (C.37) and (C.45) define systems that solve for 6 unknowns, $X := \{\tilde{u}, \tilde{u}', \tilde{v}, \tilde{v}', \tilde{w}, \tilde{p}\}$. In matrix form, this reads,

$$\begin{aligned} \frac{\partial X_i}{\partial s} &= A_{ij} X_j + F_i \quad \text{in the upper layer, and} \\ \frac{\partial X_i}{\partial t} &= A_{ij} X_j + F_i \quad \text{near the surface,} \end{aligned} \quad (\text{C.46})$$

where in the upper layer $\mathbf{A}$ is defined by eq. (C.37) as

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 + \frac{U^{0*}}{K^*} i \cos \beta & -\frac{1}{K^*} \frac{\partial K^*}{\partial s} & 0 & 0 & \frac{1}{K^{*2}} - \frac{1}{K^*} \frac{\partial K^*}{\partial s} i \cos \beta & \frac{1}{K^*} i \cos \beta \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 + \frac{U^{0*}}{K^*} i \cos \beta & -\frac{1}{K^*} \frac{\partial K^*}{\partial s} & -\frac{1}{K^*} \frac{\partial K^*}{\partial s} i \sin \beta & \frac{1}{K^*} i \sin \beta \\ -i \cos \beta & 0 & -i \sin \beta & 0 & 0 & 0 \\ -2i \cos \beta \frac{\partial K^*}{\partial s} & -i K^* \cos \beta & -2i \sin \beta \frac{\partial K^*}{\partial s} & -i K^* \sin \beta & -K^* - U^{0*} i \cos \beta & 0 \end{bmatrix}, \quad (\text{C.47})$$

and $\mathbf{F}$ is defined as

$$\mathbf{F} = \begin{bmatrix} 0 \\ -\frac{1}{K^*}\phi_n \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (\text{C.48})$$

and near the surface, $\mathbf{A}$ is defined by eq. (C.45) as

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{K^{*2}}{\kappa^2} + \frac{K^*U^{0*}}{\kappa^2}i\cos\beta & 0 & 0 & 0 & \frac{1}{\kappa^2} - \frac{1}{\kappa}\frac{\partial K^*}{\partial t}i\cos\beta & \frac{K^*}{\kappa^2}i\cos\beta \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{K^{*2}}{\kappa^2} + \frac{K^*U^{0*}}{\kappa^2}i\cos\beta & 0 & -\frac{1}{\kappa}\frac{\partial K^*}{\partial t}i\sin\beta & \frac{K^*}{\kappa^2}i\sin\beta \\ -\frac{K^*}{\kappa}i\cos\beta & 0 & -\frac{K^*}{\kappa}i\sin\beta & 0 & 0 & 0 \\ -2i\cos\beta\frac{\partial K^*}{\partial t} & -iK^*\cos\beta & -2i\sin\beta\frac{\partial K^*}{\partial t} & -iK^*\sin\beta & -\frac{K^{*2}}{\kappa} - \frac{K^*}{\kappa}U^{0*}i\cos\beta & 0 \end{bmatrix}, \quad (\text{C.49})$$

and $\mathbf{F}$ as

$$\mathbf{F} = \begin{bmatrix} 0 \\ -\frac{K^*}{\kappa^2}\phi_n \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \quad (\text{C.50})$$

Note that whilst (C.46) is written as a partial differential equation, this is purely a matter of consistency. Since the equations are solved by parametrising k and β such that s (or t) is the only variable, the partial derivative could equivalently be written as an ordinary derivative. This system of ordinary differential equations can be solved numerically. Details of the solution can be found in appendix D.